Publication strategy and pre-submission risk register

Version 2026-11-01 (first version 2026-10-29), for Raz Vakil. Companion to `paper/clc.tex`.

CLC number: a squarefree composite n that is both Carmichael ((p−1) | (n−1) for every prime p | n) and Lucas–Carmichael ((p+1) | (n+1) for every p | n). The same objects appear in the literature as **1-Williams numbers** (see §5; the literature comparison is being completed and governs the final wording of every novelty claim).

1. What we have

#	result	type	status
1	**Theorem A** (= McIntosh–Dipra 2015, Thms 2–3, d = 4; restated with an independent proof). All $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{(2p_1p_2p_3)}$, $p_2^2 \le 24p_1^3 + 1$.	known theorem, our proof	**not new** (established 2026-11-01 from the full text); kept as a self-contained restatement
2	**Theorem B.** No four-prime Williams number with $n \le 10^{30}$; none with $p_1 \le 500{,}000$ (any n). Extends McIntosh–Dipra's $p_1 > 2\times10^4$, $N > 10^{24}$.	computation	**certified** 2026-11-01 (`clc/k4/CERTIFY_*.txt`): full coverage, all survivors re-tested in Python
3	No Williams number below 10^{24} (factor-and-test of the complete Carmichael tabulations; integrity checks; full match histogram: 14 four-match numbers, 826 three-match).	computation on external data	done twice; second run certified
4	Synthesis of the Lucas–Carmichael / Baillie–PSW / Agrawal / Howe connections.	exposition	wording depends on §5
5	Polynomial-family impossibility theorems; exhaustive strata (outputs being regenerated into `agrawal/results/`).	theorems + searches	Appendix A
6	Subset-product margin measurement (exploratory, with limitations); four-prime heuristic (one moderated paragraph in §8). **Onset model withdrawn 2026-11-02.**	speculative	`paper/supplement.pdf` v2

Structure after the second referee review (2026-11-01): Introduction with a comparison table against McIntosh–Dipra → Preliminaries → structural bounds (Theorem A, attributed) → algorithm (displayed pseudocode, reconstruction lemma) → completeness/validation/ certification → results → the 10^{24} census (input data / per-entry verification / integrity checks) → discussion → Appendix A (polynomial families only). The subset-product measurement and the onset model are now a separate supplementary note (`paper/supplement.tex`). Title: *Exhaustive searches for Williams numbers: the four-prime case and a general 10^{24} bound*. AI use is declared in a separate section before the references (Elsevier/JNT format).

2. Venues

The journals' AI policies decide eligibility; the assessments below are from the external review of 2026-11-01 and must be rechecked against the live policy pages immediately before submission.

venue	fit	eligibility / reservation	recommendation
arXiv (math.NT)	natural first step	permits disclosed AI assistance; a first-time submitter may need an endorsement	submit after Gates A–B
Canadian Mathematical Bulletin	good (Williams's question appeared there)	Cambridge requires declared AI use and full human accountability	**primary journal target**; ask the editor in advance about substantive AI involvement in proofs and code
Experimental Mathematics	good (computation + heuristics)	Taylor & Francis forbids generative-AI first drafts; detailed declaration required	possible only with a human-rewritten manuscript
Journal of Number Theory	stretch	plausible only if the inequality and the search advance are clearly substantial beyond McIntosh–Dipra (JNT 2015)	decide after §5
Mathematics of Computation	aspirational	needs algorithmic or certification novelty; endorsement from a consultant would matter	decide after §5
Journal of Integer Sequences	constrained	forbids LLM-written English; requires detailed credit for proof suggestions; discourages brute-force term extensions	only after a genuinely independent human rewrite
Integers	**ineligible**	its policy states it will not consider work using AI to produce mathematics, code or text	removed

OEIS: concise comments on A006972 / A002997 (and the near-miss data on A329948) only after the preprint identifier exists; OEIS is a record, not a priority mechanism. Code and owned data: Zenodo deposit with a versioned DOI; third-party inputs are cited and checksummed (`clc/DATA_SHA256.txt`), not redistributed.

3. Dissemination order

1. Finish the certification (Gate A), the literature comparison (§5) and the human proof review.
2. Send consultants the manuscript first, with assigned questions (proof correctness; enumeration completeness; prior art; venue level). Share the AI audit reports afterwards, so they do not anchor the first reading.
3. Freeze a release candidate: tag, archive code, logs, hashes and result files under a versioned DOI.
4. Submit a policy-compliant, human-accountable preprint to arXiv.
5. Post the OEIS comments, citing the arXiv identifier.

6. Address substantive feedback; submit to the chosen journal when the completion gates below are met, not after a fixed elapsed time.

4. Authorship and AI disclosure

The AI system (Claude, Anthropic) was used for proof discovery and drafting, code generation, prose drafting, literature search, and adversarial checking. The disclosure must name these roles separately (the current acknowledgement does). The author must independently reconstruct and accept responsibility for every proof, computation, citation and sentence; the two AI audit reports in `docs/` are adversarial checks, not independent human review, and are labelled accordingly. A consultant is offered co-authorship only for a contribution meeting ordinary authorship standards.

5. Literature and novelty (final)

Full texts of McIntosh–Dipra (JNT 2015) and Zhang (Math. Comp. 2015) were read on 2026-11-01 (`docs/literature-williams-numbers.md`). **Theorem A is not new**: it is McIntosh–Dipra's Theorems 2–3 for d = 4 (they even have the better $p_3 < p_1 p_2$ and prove $\Pi < 1$ outright). Lemma 2.2(c) and the k = 3 proof are theirs as well. What the manuscript still contributes:

1. **Theorem B** — the four-prime exclusion extended from ($p_1 > 2\times10^4$, $N > 10^{24}$) to ($p_1 > 500{,}000$, $n > 10^{30}$), by a different and faster algorithm (CRT class for the cofactor + window filter), with a certified completeness record.
2. **The 10^{24} all-k verification** with completeness/OEIS cross-checks and near-miss statistics (McIntosh–Dipra's all-d search states no per-d bound).
3. A self-contained exposition linking the Williams-number, Baillie–PSW, Agrawal and Howe literatures (with correct citations now).
4. Appendix material: polynomial-family theorems, two-sided subset-product measurements compared with Zhang's Conjecture 4.

Assessment. This is now a short computational note, not a theorem paper. Realistic outlets: arXiv (certainly worth posting as a record of the computation and the algorithm); a computational journal only if the referee values the $10^6\times$ extension and the certified method — Mathematics of Computation is unlikely, CMB possible but a stretch, JIS/Integers blocked by AI policy. An alternative with real value at low cost: send the results to McIntosh (and Zhang), who will know whether a note is warranted, and post the OEIS comments. **The author should decide whether to proceed to a journal at all**; the arXiv posting and OEIS record are recommended regardless.

6. Risk register (neutral form)

risk	mitigation in place	still to do
Proof correctness (Theorem A)	two reconstructions; executable checks of every identity (`tests/test_theorems.py`); two AI audits, one of which found and repaired a false estimate in the $\Pi > 1$ case	human number-theorist review of §3

risk	mitigation in place	still to do
Enumeration completeness (Theorem B)	lemmas stated and proved; completeness proposition in §3; both candidate walks agree on six instances; relaxed-problem regression to 10^{12} archived; per-block logs; **certified** by `certify_k4.py` (`CERTIFY_*.txt`)	—
Verifier soundness	verifier checks the original tuple without normalisation; negative tests; C Fermat routine cross-checked against Python	—
Dependence on external tabulations (10^{24})	three completeness checks; per-ω cross-checks against two OEIS sequences; checksums	—
Prior art	literature note written; manuscript reframed	read McIntosh–Dipra and Zhang in full; resolve the two TODOs
AI policy compliance	roles disclosed; venue matrix	recheck policies at submission; human-authored prose where required
Process errors	the rerun's last blocks were lost to a `git stash -u` on live log files (2026-11-01) and recomputed; the wrapper now fails fast and certifies coverage	never run file-removing git commands while a job writes into the tree

6a. Second referee review (2026-11-01, `docs/review-referee-2026-11-01.pdf`)

Major revision with a favourable view; both certifications repeated by the reviewer from the clean tag. All requested items applied the same day: 'first attained at' → 'attained, for example, at' (the exact four-match count follows from the running rescan); displayed pseudocode + reconstruction lemma + completeness proposition placed after the algorithm; certificate asserts per-block survivor counts and Fermat counts; Pomerance claim narrowed to 'any counterexample produced by the construction'; heuristic moderated ($24/(p^2-1)$, 'consistent with'); Appendix A qualified (identities in k; gcd normalisation explained; outputs archived); subset-product and onset material moved to a supplement; section order as recommended; precise inventory of what was rerun / cross-checked / single-run; 'integrity checks' wording; MR-splitting sentence clarified; rescan worker errors fatal + rescan certifier; survivors 'deduplicated within campaigns'; shorter title; concise abstract; internal audit chronology removed from the paper; MSC/keywords/PDF metadata; DOIs added. Not changed: Pinch 2007 page range (reviewer says 129–131, our fact-check found 113–128 via OEIS) — **verify against the TUCS volume before submission**.

6b. Third review (Codex, 2026-11-02, `docs/review-codex-2026-11-02.pdf`)

Main paper "close to submission-ready". Applied: final tag `v1.0` cited in the paper and certificates regenerated from the clean tag (binary rebuilt from tagged source; reports record tag, source/binary hashes, no dirty tracked files); AI declaration now names both systems (Claude: authoring/code/checks; OpenAI Codex: three review rounds, literature discovery, repeated

certifications) and states that no independent human review had been completed; Algorithm 1: M* condition and the +1/+3 bound made explicit; Reproducibility is a numbered section; PDF metadata title fixed; Table 1 caption broadened; date updated. **Supplement: the quantitative onset estimate (10^{208}–10^{490}, k ≈ 80–110) was found to be internally inconsistent and is WITHDRAWN**; the note is now an exploratory report of the subset-product measurement with an explicit limitations section, and the main paper cites it only as unpublished material on which nothing depends. `docs/obstructions.md` §6 carries the same withdrawal notice.

7. Gates

Gate A — before any public posting - [ ] Theorem A proof text final (both Π branches explicit; δ sign correct); reviewed by a human number theorist. - [x] `CERTIFY_n1e30.txt` and `CERTIFY_p1_500000.txt` present, ending in CERTIFIED (3,811/3,811 and 323/323 blocks; 1,904 and 1,185 distinct survivors, all re-tested in Python) — 2026-11-01 13:04. - [x] Literature comparison written; novelty claims rewritten against McIntosh–Dipra and Zhang. - [x] McIntosh–Dipra and Zhang read in full (2026-11-01); the two TODOs in the paper resolved; Theorem A reattributed. - [x] Implementation-completeness proposition written into §3 (Proposition 3.6).

Gate B — release candidate - [x] `./run_tests.sh` passed from a clean archive of v1.0-rc5 on a fresh EC2 c7i.2xlarge (Ubuntu 24.04, gcc, fresh venv), 2026-11-02; transcript `docs/clean-clone-test-2026-11-02.txt`. - [x] Source frozen at tag `v1.0` (commit b2af6b3). The certificates were generated from commit f5e2e50, whose source is identical to `v1.0` except for the certificate files themselves (a certificate cannot be inside the commit it names); they record the SHA-256 of the enumerator source and binary, which are the same at both commits. GitHub release `v1.0` carries the source archive. - [ ] Zenodo DOI (needs the repository public or a Zenodo token; author action). - [x] 10^{24} histogram rescan finished and certified 2026-11-02 (`clc/CERTIFY_1e24_rescan.txt`): exactly 14 four-match numbers; inserted in §7. - [ ] Family-search outputs regenerated (`agrawal/results/`, m=4 a≤500 and m=5 a≤150 running 2026-11-01 afternoon). - [x] Pinch 2007 page range: resolved 2026-11-02. The referee's 129–131 is Pinch's *other* paper in TUCS GP 46, "The Carmichael numbers up to 10^{21}" (so cited in Baillie–Fiori–Wagstaff 2021's references); "Absolute quadratic pseudoprimes" is pp. 113–128 (OEIS A392556/A392557). Both now in the bibliography. Final visual check possible from the volume PDF: https://tucs.fi/publications/attachment.php?fname=G46.pdf (site certificate chain incomplete; a browser may warn). - [ ] Every bibliography entry verified against the primary source.

Gate C — journal submission - [ ] Target journal's AI policy confirmed against the actual use history (written guidance if ambiguous). - [ ] Prose human-authored to the standard the venue requires. - [ ] Disclosure text matches the roles in §4. - [ ] Substantive consultant feedback resolved.